Chain Surveying

Linear Measurements

1) Direct methods - chain, tape

2) Optical methods - theodolite/tacheometer

3) Electronic methods - EDM, total Station

Approximate methods

1) Eye Judgement

2) Pacing – Degree of accuracy is 1 in 100.

{Distance = No of Steps × Average pace length (80cm)}

3) Passometer → Counts the no of pases. (size of a Watch)

4) Pedometer Displays → Displays distance travelled by Calculation (modified passometer)

5) Perambulator or Surveyor's Wheel or way wiser → It consists of a handle a rod a wheel and a dial.

6) odometer → It is an equipment attached to wheel of a vehicle that counts the no of revolutions of wheel and displays the distance travelled.

7) Speedometer → Distance = Time × Velocity.

Chain Survey

preferred in:

→ Fairly level ground

→ Small areas

→ Open areas with less no of details

Principle of Chain Survey:

→Triangulation

→ Triangles Should be well Conditioned

Well Conditioned triangle

→No angle <30° or 120°

Ill Conditioned triangle

→At least one angle <30° or> 120°.

Best Conditioned triangle

→Isosceles triangle having angles 56°14 each OR Equilateral triangle.

Instruments and Accessories

1. Chain
2. Tape
3. Peg
4. Arrow
5. Lath and Whites.
6. Ranging rod.
7. Ranging Pole
8. offset rod
9. Plumb bob
10. Line Ranger
11. Cross Staff/optical Square

1. Chain

→ Brass handle provided with groove to hold the chain close to Arrow.

→ Handle is connected to chain by Swivel joint/ I bolt connection.

→ Chain is maid up of Galvanised mild Steel 4 mm thick or (8 swg)

→ 3 circular rings and swivel joint helps in flexibility of chain.

Types of chain

1) Metric chain (m)

2) Non metric Chain (ft)

1) Metric Chain (m)

→ IS code for metric chain – IS 1492 (1970).

→ Length 5, 10, 20, 30

→ Widely used

→ link = 20 cm. (1m = 5 links)

→ Brass rings at every meter

→ tally at every 5m

2) Non-metric Chain

1) Engineer's chain

100 ft length → 100 links → 1 ft/ link

2) Surveyor's chain or Gunter's chain

66 ft long → 100 links → 0.66 ft/ling Link

I furlong = 660ft = 10 Surveyor's Chain

I mile = 8 furlong = 80 surveyors Chain

3) Revenue Chain

33ft long → 16 links = 33/16= 2 1/16 ft/link .

→ Used in cadastral surveying.

Testing of chain

→ Stretched at 8 kg Pull at 20°C and tasted using Steel tape.

Chain length	Permissible error
1m	±2mm
5m	± 3mm
20 m	±5mm
30m	± 8mm

<u>Field work</u>

2 people → 1 leader and 1 follower.

→ follower throws Chain.

→ Leader fixes arrow in ground.

→ Follower removes arrow from ground.

→ Follower gives instruction to leader.

→ Leader obeys instructions given by follower.

→ Leader Stretches chain.

<u>2.Tape</u>

Types:

<u>1. Cloth or linen tape</u>

→ Less durable.

→ Affected by moisture.

→ 20 or 30m maximum.

→12 to 15 mm wide.

<u>2. Metallic tape</u>

→ Wires of Copper or brass interwoven with cloth/linen.

→ Max 50m.

3. Steel tape

→ Made up of steel

→ Better Accuracy

→ Testing of chain

→ 50m Max

→ 6 to 10 mm wide

→ Provided with a winding device

4. Invar tape

→ Made up of Invar - Alloy of NIKEL (36%) and STEEL (64%).

→ Very low coefficient of thermal expansion (α).

→ α Invar = 0.5×10^{-6} / °F
= 0.9×10^{-6}/ °C

→ α Steel = 12×10^{-6}/ °C

→ α concrete = 10×10^{-6}/ °C

→ Very Accurate

→ Base line measurements

Condition	**Permissible error/ Accuracy**
1) Rough measurement with Chain on hilly terrain.	1 in 250
2) Using chain on average Conditions of ground.	1 in 500
3) Careful measurement with Chain in level ground	1 in 1000 (max chain)

4) Rough measurement with tape	1 in 1000
5) Careful measurement with tape	1 in 2000
6) Measurement with invar tape.	1 in 10,000

<u>3.Wooden peg</u>

→ Used for marking Stations

→ Top cross section – 2.5 × 2.5 cm.

→ Length - 15cm.

→Above land 4 cm.

<u>4.Arrows</u>

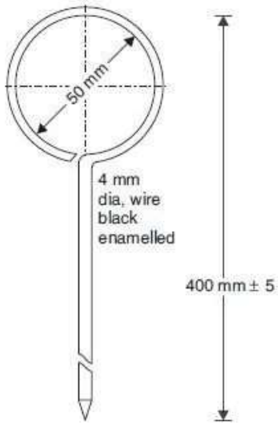


Fig. 12.5. Arrows

→ To mark end of chain length.

→ 10 arrows/chain

→ Material - hardened and tempered mild Steel 4mm thick (8 SWG)

5. Lath and Whites

→ To fix intermediate Points.

→ 0.5m to 1m long Soft wood piece.

6. Ranging rod

→ For ranging and Sighting points.

→ Red and White or black and White bands.

→ 20cm long bands.

→ Circular Cross Section 3cm diagonal.

→ 2m or 3m long.

→ Metal Shoe is provided.

→ Range of visibility with naked eye - 200m.

7. Ranging Pole

→ 4 to 8m long

→ 6 to 10 cm diameter.

→ 20 cm bands.

8. offset rod

→ To measure Short offsets.

→ A hook or notch at top to move chain.

Field book

It is an oblong book of size

→ 12 cm by 20cm in which the measurement from field are recorded.

Types of field book

1. single line field book.

2. Double line field book.

1) Single line field book

→ It consists of a Single central red line.

→ Important detailed and large scale.

<u>2) Double line field book</u>

→ Two blue lines Spaced 1.5 to 2 cm apart.

→ Ordinary Works.

<u>offset</u>

It is the distance from a Survey line to a detail.

Types

→ Long offset > 15m

→ Short offset < 15m (can be set out by eye)

→ Perpendicular offset – 90°

→ Oblique offset not 90°

<u>Setting out offset</u>

<u>1) By wing chain</u>.

→ By Swinging chain (min distance will be in 90°) By 3-4-5 or 6-8-10 rule

(Pythagoras triplets)

<u>2) Using Cross staff</u>

1. open cross staff

→ To erect perpendicular offsets (90°).

→ Height = 1.2m.

2. French Cross staff

→ Octagonal shaped

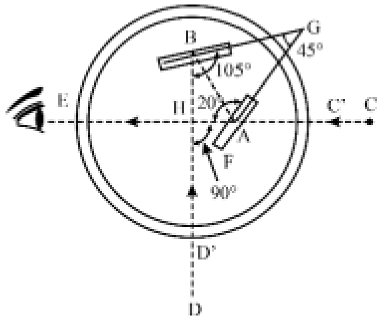
→ To erect offsets of multiples of 45°.(45°, 90, 135°, 180°... etc).

→ Distance between opposite slit of French cross staff – 8cm.

3. Adjustable Cron staff

→ Any angle can be taken.

<u>3) using optical square</u>



→ 2 mirrors.

→ H-Horizon mirror-half Silvered

→ I - Index mirror - fully Silvered.

→ Principle double reflection

→ Angle between horizon mirror Index mirror = 45°.

→ Angle between Incident ray and reflected ray = 90°.

→ Angle between horizon mirror and horizon Sight = 120°.

→ Angle between Index mirror and Index Sight= 105°.

→ Better Accuracy than cross Staff and Chain.

→ Only perpendicular offsets.

4) using Prism Square

→ Using Pentaprism (5 faces).

→ Principle Double reflection.

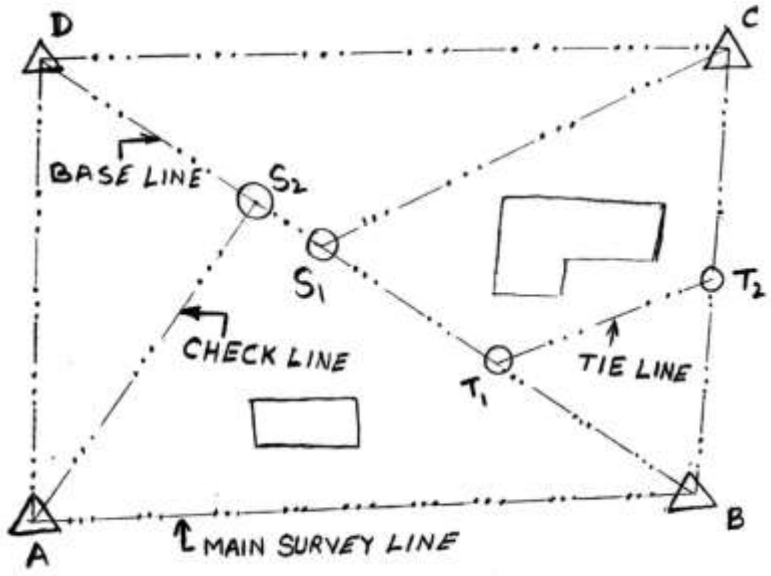
→ Best Accuracy (greater than optical square)

→ Angles between reflecting faces in a prism Square = 45°

5) Using Sight Square

→ It consists of two telescopes Placed at right angles.

Terms in chaining



Subsidiary Station / tie Station

It is a Station Selected on main Survey line to reduce length of long offsets. (To plot interior details).

Subsidiary line/tie line

A line Joining Subsidiary Station to reduce long offsets.

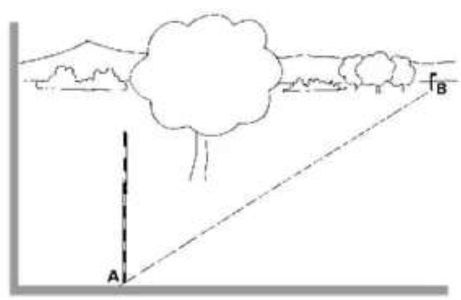
Check line (To check Accuracy)

It usually connects corner of a triangle to a point on opposite side.

→ Subsidiary lines/tie lines Can be used as check lines.

→ Check line is also known as proof line.

Ranging



→ Process of fixing intermediate Points.

Types of Ranging

→ Direct ranging (Intervisible)

→ Indirect ranging (not intervisible)

Direct ranging

→ Ranging by eye - 3 Ranging rod 2 person.

→ Ranging by line ranger- (Reflecting Instrument Angle between reflecting faces 90°) (2RR, 1 person)

→ Direct ranging by theodolite - lining in (2 RR, 2 person)

Indirect ranging

→ Reciprocal ranging - 4RR

Used in : Hill , Far (ranging rod visibility up to 200 m).

→ Random line method - To cross wooded areas.

→ Indirect ranging by theodolite - Balancing in.

Q) Minimum no. of ranging rods required for direct ranging?

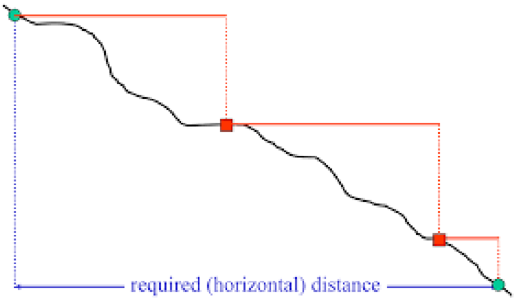
Ans: 3 Ranging rod.

Measurement in Sloping ground

1) Direct method.

2) Indirect method.

1) Direct method (Stepping)



→ D=L1+L2 +L3

→ Method of stepping.

→ Plumb bob or drop arrow.

→ Preferred downhill.

2) Indirect method

→ Sloping distance is measured first from which horizontal distance is calculated.

A) Sloping distance and Slope known.

→ Angle (θ) is measured by clinometer.

→ Distance (D) = Sloping distance (L) cos θ.

$$D = L \cos\theta.$$

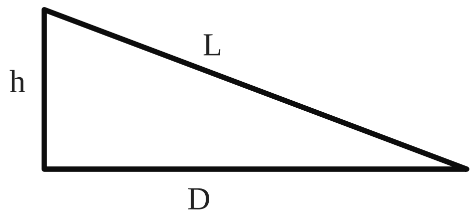
→ Error = Measured- actual
= L - D
= L - L cos θ = L(1 - cos θ)(+ ve)

→ Correction for slope = - L (1 – cos θ)

→ Correction for slope is always (-ve).

B) Sloping distance and level difference known

$D = \sqrt{L^2 - h^2}$



Error = L - D
$= L - \sqrt{L^2 - h^2}$

$= \dfrac{h^2}{2L}$

Therefore, Slope Correction = $-\dfrac{h^2}{2L}$

C) Hypotenusal Allowance.

→The length difference between horizontal line and hypotenuse is called as HA (Hypotenusal allowance)

→ In hypotenusal allowance the Correction are applied at the field itself.

→ HA = $\frac{L}{2}\theta^2$ meters = 50 θ^2 links

θ in radian

= $\frac{L}{2n^2}$ meters = $\frac{50}{n^2}$ links

→ HA = $L(sec\,\theta - 1)$ meters
= $100(sec\,\theta - 1)$ links

θ in degrees

HA = 0.015 θ^2 links (only equation to do numerical)

Q 10° Slope ground HA?

HA = 0.015 θ^2 links
= 0.015 × 100 = 1.5 links

Obstructions in chaining

1). Obstructions to both chaining and ranging

eg- buildings.

2). Obstruction to chaining but not ranging (vision).

eg- Water bodies.

3). Obstruction to ranging (vision) but not Chaining.

eg- Hill, Far.

ERRORS

Based on Source

1) Instrumental error - Permanent adjustments

2) Personal error -Temporary adjustments.

3) Natural error - Temperature, pressure, wind Vibration, humidity.

Based on nature

1) Mistakes - Can be avoided by care (reading , recording)

2) Systematic errors

3) Accidental errors.

Systematic errors

→ It follows a definite law or theorem.

→ Using which a correction can be calculated and applied.

→Under Same Conditions systematic errors have Same size and sign.

→ Cumulative errors Proportional to L

Accidental errors

→ Errors which remains after eliminating Mistakes and Systematic errors.

→Occur in all measurements

→ It follows law of probability

→Compensating error $\alpha \sqrt{L}$

Systematic errors (Tape Corrections)

1. Correction for Standardization

→ Correct Length = measured length × $\frac{\text{Actual length of chain}}{\text{Standard length of Chain}}$

→ Correct area = measured area × $\left\{\frac{\text{Actual length of chain}}{\text{Standard length of Chain}}\right\}^2$

Long Chain = Measured L < Actual L
→ Error (-ve)
→ Correction (+ve.)

Short Chain = Measured L > Actual L
→ error (+ve)
→ Correction (-ve)

2. Correction for temperature

C temp = L $\alpha\ (T_m - T_0)$
Where,
L-length measured

T_m - Mean field tempreture

T_0 - Standard tempreture

α - Coefficient of termal expansion.

Tm > To

long chain : -ve Cumulative error , Correction +ve

3. Correction for Pull

$$C\ pull = \frac{(P - Po)\ L}{AE}$$

Where,
P- Pull applied

Po - Standard pull

L - length measured.

A – Area of cross section.

E – Youngs modulus of elasticity of material.

P> Po

Chain long : Error -ve Correction +ve.

→ Correction for pull can be considered both Compensating and cumulative as applied.

4. Correction for sag

→ Due to Sag chain is always short
→ Error +ve, Correction -ve (Always)

$$C\ sag = \frac{w^3 L^3}{24n^2P^2} = \frac{W^3 L}{24n^2P^2}$$

Where,
w= Weight/m of chain.

W = total weight of chain.

L-length of chain.

P- Pull applied.

n - no. of sag.

.

Normal pull

The pull applied at which Correction for sag equal to Correction for pull or effect of pull balances effect of sag.

5. Slope Correction

C slope = - L (1 – cos θ) (Angle)

$= -\frac{h^2}{2L}$ (height)

→ Correction for Slope is always negative and error is Positive.

6. Correction for bad ranging.

C ranging = $-\frac{d^2}{2L}$

→ Error due to bad ranging is always Positive.